
Redyx — User Service Test Cases

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User Service — Test Case Specification

Project Name	Redyx
Module Name	User
File	internal/user/server_test.go
Total Tests	11
Package	user (same-package test)
Created By	idityaGE
Date of Creation	2026-05-10

Test Infrastructure

Component	Implementation
Logger	<code>zap.NewNop()</code>
DB Pool	<code>nil</code> — tests short-circuit before DB access
Post / Comment / Community Clients	<code>nil</code> — graceful nil-client behaviour tested explicitly
Auth Context	<code>auth.WithClaims(context.Background(), &auth.Claims{...})</code>
Helper convention	<code>t.Helper()</code> in <code>testUserServer</code> , <code>authedUserCtx</code>

1. profileToProto Conversion Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_001	TestProfileToProto_ActiveUser		Construct active profile struct; call <code>profileToProto</code>	<code>userId = "user-abc", username = "alice", displayName = "Alice Wonderland", bio = "Curiouser and curiouser", karma = 1234</code>	All fields mapped correctly: <code>UserId</code> , <code>Username</code> , <code>DisplayName</code> , <code>Bio</code> , <code>Karma</code> all match source	PASS

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_002	TestProfileToP DeletedUser_- Sanitized	Profile with Delete- dAt.Valid=1	Set Delete- dAt.Valid = true; call profileToProto	username = "deleted_ user", displayName = "Secret Name", bio = "Private bio", karma = 9999	Username="[del Display- Name="", Bio="", Karma=0 (all PII stripped for soft-deleted accounts)	PASS

2. GetProfile Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_003	TestGetProfile_ User server EmptyUsername	initialized	Call GetProfile with empty username	username = ""	Returns an error	PASS

3. UpdateProfile Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_004	TestUpdateProfile_ Unauthenticated	User server initialized	Call UpdateProfile with unauthenticated context	displayName = "New Name"	Returns an error (missing auth claims)	PASS
TC_USER_005	TestUpdateProfile_ BioTooLong	Authenticated context	Call UpdateProfile with 501-character bio	bio = 501×"a"	Returns an error (max 500 chars)	PASS
TC_USER_006	TestUpdateProfile_ DisplayName- TooLong	Authenticated context	Call UpdateProfile with 51-character display name	displayName = 51×"a"	Returns an error (max 50 chars)	PASS

4. DeleteAccount Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_007	TestDeleteAccountUnauthenticated	User server initialized	Call DeleteAccount with unauthenticated context	—	Returns an error	PASS

5. GetUserPosts Tests (Nil-Client Graceful Behaviour)

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_008	TestGetUserPostsEmptyUsername	User server initialized	Call GetUserPosts with empty username	username = ""	Returns an error	PASS
TC_USER_009	TestGetUserPostsNoPost-Client_ReturnsEmpty	User server with nil post client	Call GetUserPosts with valid username	username = "someuser"	Returns success with empty posts list (graceful nil-client fallback)	PASS

6. GetUserComments Tests (Nil-Client Graceful Behaviour)

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_010	TestGetUserCommentsEmptyUsername	User server initialized	Call GetUserComments with empty username	username = ""	Returns an error	PASS
TC_USER_011	TestGetUserCommentsNoComment-Client_ReturnsEmpty	User server with nil comment client	Call GetUserComments with valid username	username = "someuser"	Returns success with empty comments list (graceful nil-client fallback)	PASS

7. GetUserCommunities Tests (Nil-Client Graceful Behaviour)

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_USER_012	TestGetUserCommunities - EmptyUserID	User server initialized	Call GetUserCommunities with empty user_id	userId = ""	Returns an error	PASS
TC_USER_013	TestGetUserCommunities - NoCommunity-Client_ReturnsEmpty	User server with nil community client	Call GetUserCommunities with valid user_id and pagination	userId = "user-123", limit = 10	Returns success with empty communities list (graceful nil-client fallback)	PASS

Patterns & Conventions

Pattern	Implementation
Package scope	package user (white-box / same-package)
Logger	zap.NewNop() (silent)
Nil DB	NewServer(nil, zap.NewNop()) — validation-only tests
Nil client graceful fallback	GetUserPosts / GetUserComments / GetUserCommunities return empty response when downstream client is nil (no panic)
Soft-delete sanitization	profileToProto clears PII fields when DeletedAt.Valid=true
Auth context	auth.WithClaims(context.Background(), &auth.Claims{UserID, Username})
Helper tagging	t.Helper() in testUserServer and authedUserCtx